Recursion and Call Stacks

Recursion is a programming technique where a function solves a problem by calling itself.
Two Essential Components of Every Recursive Function:
1. Base Case: The terminating condition that produces a result directly without further recursion.
Without a base case, recursion leads to infinite calls causing a Stack Overflow error.
2. Recursive Step: The rule that reduces the original input towards the base case.
Call Stack Mechanics: Each recursive call pushes a new stack frame containing local variables and return address.
Once a base case is hit, stack frames unwind in Last-In-First-Out (LIFO) order.
Common Socratic Pitfalls: Students often forget to return the result of the recursive call,
or fail to make progress toward the base case with each step.